

Osman Can Kandemiroglu

Marine molecular biogeochemist — mass spectrometry imaging, lipid & glyco-biosignatures

APPLICATION

SFB 1759 · subproject C01

Enceladus · praedoc

Ref. SFB1759-C01-NK

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Certificates: English C1 (UniCert III) · German C1 (Goethe-Zertifikat, TestDaF)

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Marine molecular biogeochemist and geophysical engineer who combines **high-resolution organic mass spectrometry** (FT-ICR-MS, MALDI/LDI imaging, GC-MS, HPLC-MS) and lipid-biomarker analysis with a **hydrothermal-systems background** and hands-on **experimental instrument-building**. I am eager to learn **LILBID and impact-ionization mass spectrometry** to detect lipids and fatty acids in ice grains and to trace organics from **Enceladus's hydrothermal vent systems** into the plume — the aim of **SFB 1759 subproject C01** (Enceladus). As a trained geophysicist (seismic-signal interpretation) and an engineer who builds experimental systems from scratch, I am as comfortable on the instrument and physics side as on the molecular side.

CORE METHODS & TECHNIQUES

High-resolution mass spectrometry

FT-ICR-MS; MALDI/LDI-FT-ICR MS imaging (Bruker solarix XR); GC-MS; HPLC-MS (APCI); GC-IRMS / compound-specific stable-isotope analysis

Lipid-biomarker & organic geochemistry

Total lipid extraction (ultrasonication, centrifugation); open-column liquid chromatography; derivatisation / silylation; hydrocarbons, sterols, GDGTs, intact polar lipids, hopanoids, alkenones, pigments / carotenoids, quinones

Aqueous & instrumental (earlier training)

Aqueous geochemical modelling (PhreeqC, AquaChem); **ICP-MS**, GF-AAS, FE-SEM-EDX; borehole geophysical logging (full-waveform sonic, acoustic televiewer, gamma-ray)

Experimental & cultivation biogeochemistry

Marine bacterial cultivation (*Vibrio* spp.); long-term & anaerobic incubations; ¹³C stable-isotope probing; alginate model-particle synthesis (emulsification / internal gelation); **building experimental set-ups from scratch**

Microscale imaging & geochemical mapping

MS imaging (molecular mapping); μ -XRF elemental mapping; microsensor profiling (O₂, H₂S, pH, redox); time-lapse microscopy

RESEARCH & PROFESSIONAL EXPERIENCE

Oct 2024 –

Dec 2025

Master's Researcher — Organic Geochemistry & Marine Glycobiology

AG Hinrichs (MARUM) & AG Hehemann (MPI–Marine Microbiology), University of Bremen

Designed and built a laboratory model of the marine carbon cycle from synthetic alginate microparticles and marine *Vibrio* bacteria, and tracked particle degradation in real time by time-lapse microscopy. Thesis: *"Instant visualisation of organic matter transformation in the ocean."* Supervisors: Prof. K.-U. Hinrichs, Prof. J.-H. Hehemann.

Aug 2023 –

Sep 2024

Researcher — Geoscientific Research Project

AG Hinrichs, MARUM, University of Bremen

Mapped microscale molecular distributions across redox zones by LDI/MALDI-FT-ICR-MS imaging integrated with μ -XRF and microsensor profiling; identified pigments (Chl-*a*), quinones (PQ9:9) and candidate far-red-light-adapted cyanobacteria. Framed explicitly as an **early-Earth / Mars analogue and biosignature-detection** study. Project: *"Light-driven structuring of a hypersaline microbial mat (Mallorca)."* Since graduating, developing software to read and process LDI/MALDI-FT-ICR-MS imaging datasets (in preparation, 2026).

May 2022 –

Jan 2023

Research Assistant — Organic Geochemistry Laboratory

Alfred-Wegener-Institut (AWI), Marine Geology & Palaeontology

Ran bulk and biomarker organic-geochemical analysis of sediments from IODP Expedition 374 and cruises PS81 and PS119 — total lipid extraction, open-column chromatography and silylation, followed by GC-MS and HPLC-MS analysis and interpretation.

2016 – 2018

Graduate Researcher (Hydrogeochemistry) & Teaching Assistant

Ruhr-University Bochum, Applied Geology / Hydrogeology

Investigated trace-element and organic-pollutant behaviour in groundwater and drinking-water treatment, with hydrogeochemical modelling in PhreeqC / AquaChem. Taught inorganic and organic hydrochemistry; led field courses, including a 10-day geothermal / hydrogeology course in Argentina; supervised an MSc thesis and co-wrote funding proposals.

EDUCATION

2020 – 2026

M.Sc. Marine Geosciences — University of Bremen

Molecular geobiology, biogeochemistry and marine environmental archives; thesis at MARUM / MPI-Marine Microbiology.

- 2008 – 2011 **M.Sc. Applied Geology / Hydrogeology** — Ruhr-University Bochum
Thesis on the origin and water-treatment behaviour of nickel in groundwater (ICP-MS, GF-AAS, FE-SEM–EDX, PhreeqC).
- 2002 – 2006 **B.Sc. Geophysical Engineering** — University of Kocaeli, Turkey
Petroleum geophysics, seismology and engineering geology; thesis on earthquake-induced soil liquefaction.
- 2019 – 2021 **Guest Student, Geochemistry** — Utrecht University
Organic Geochemistry; Microbes & Biogeochemistry; Physical Chemistry.

PUBLICATIONS & CONFERENCE CONTRIBUTIONS

Peer-reviewed journal article

- 2012 Wisotzky, F., **Kandemiroglu, O. C.** & Plassmann, C. *Nickelfreisetzung in das Grundwasser und dessen Bindung bei der Wasseraufbereitung zu Trinkwasser (Nettetal/Niederrhein). gwf Wasser / Abwasser*, 153(7), 828–832.

Conference contributions & posters

- 2025 **Kandemiroglu, O. C.**, Gu, B. & Hehemann, J.-H. *Instant Visualization of Polysaccharide Degradation by Marine Bacteria*. Poster, Ocean Floor Symposium 2025, MARUM — Center for Marine Environmental Sciences, University of Bremen.
- 2018 **Kandemiroglu, O. C.** & Wisotzky, F. *Vorkommen von organischen Spurenstoffen (Arzneimittel, Pestizide) im Grundwasserleitern in Nordrhein-Westfalen*. Conference paper, 26th Conference of the FH-DGGV (Fachsektion Hydrogeologie in der DGGV) — "Grundwasser im Umfeld von Bergbau, Energie und urbanen Räumen", Ruhr-University Bochum, 21–24 March 2018.
- 2012 **Kandemiroglu, O. C.**, Loewe, R. & Wisotzky, F. *Hydrogeochemische Untersuchungen im Raum Nettetal zur Klärung der Nickelmobilität im Grundwasser und bei der Wasseraufbereitung*. Poster, Bochumer Grundwassertag 2012, Ruhr-University Bochum.

Technical reports & theses

- 2026 **Kandemiroglu, O. C.** *Instant Visualization of Organic Matter Transformation in the Ocean*. M.Sc. thesis, University of Bremen (MARUM / MPI-Marine Microbiology).
- 2025 **Kandemiroglu, O. C.** *Multi-proxy Evidence for Gradual Limnic–Marine Transition in the Black Sea at 8.5–8.0 ka*. Technical report.
- 2022 **Kandemiroglu, O. C.** *Coupled pore-water geochemistry and ¹³C-labelled lipid biomarkers reveal active methane-oxidizing communities across the redox zonation of the Helgoland Mud Area (German Bight, North Sea)*. Technical report.
- 2011 **Kandemiroglu, O. C.** *Untersuchungen zur Herkunft des Nickels im Rohwasser der Wasserwerke Breyell und Kaldenkirchen der Stadtwerke Nettetal GmbH sowie zum hydrochemischen Verhalten des Nickels bei der Wasseraufbereitung*. M.Sc. thesis, Ruhr-University Bochum. Full list — [ResearchGate profile](#).

FIELD, TEACHING, AWARDS & SERVICE

FIELD CAMPAIGNS & TEACHING

Near-surface geothermal drilling and multi-method borehole logging (full-waveform sonic, acoustic televiewer, gamma-ray) across 12 wells, International Geothermal Centre, Bochum; geothermal & hydrogeology field courses in Argentina and Italy (Larderello & Elba). Taught Inorganic & Organic Hydrochemistry, Introduction to Hydrogeology and Geology for Civil Engineers (RUB, 2016–2018); supervised BSc and MSc theses.

MEMBERSHIPS

Deutsche Astrobiologische Gesellschaft (DAbG, 2022); European Astrobiology Network Association (EANA, 2023); European Association of Organic Geochemists (EAOG, 2024); European Geosciences Union (EGU, 2016); The Geochemical Society (2023).

SCIENCE COMMUNICATION & SERVICE

[Personal research webpage \(osmancankandemiroglu.com\)](#), science animation, analogue photography and short film; Faculty Council member (U Bremen FB5, 2024–); [study guide \(Studienlotse\)](#) & [orientation mentor](#), Marine Geosciences (2023–); science outreach to regional schools, RUB International Office (2016–18).

REFERENCES

Dr. Juliane Müller

Alfred-Wegener-Institut (AWI), Bremerhaven
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Prof. Dr. Stefan Wöhllich

Ruhr-University Bochum, Applied Geology / Hydrogeology
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